Module 5 — Read 0

Project preview — Project 5 — Looking Ahead

Module 5, Read 0: Why prediction matters

Big question: You have a model — but how far into the future can you trust it, and what does an honest **interval** around a prediction actually mean?

Project: Project 5 — Looking Ahead · **Next:** Read 1

> ℹ Short read — start here
>
> Five-minute orientation before Read 1. Module 4 asked *"Did this treatment cause a difference?"* Module 5 asks *"Given what I measure today, what will happen tomorrow?"*

The story in one pass

In Modules 1–4 you **tested claims** and **estimated parameters**. In Module 5 you build a **predictive model**: fit a line to two numerical variables, make a **point prediction** for a new case, and wrap it in an honest **prediction interval** that reflects real uncertainty.

The core tool is **simple linear regression**: $\hat{y} = b_0 + b_1 x$. Read 1 shows you how to fit the line and measure how well it works. Read 2 shows you how to attach **confidence intervals** (for the mean) and **prediction intervals** (for one individual) to your predictions.

Project 5 uses real team data — nesting dolls or stone containers — so the prediction you report will be for an actual object your team measures.

Tour the project

Open Project 5 — Looking Ahead and identify:

Section	What to notice	Why it matters for *your* project
Option A or B	Nesting dolls (diameter → weight) or stone containers (width → weight)	Choose one option with your team
Data collection	Each team member measures the same objects independently	Reproducibility matters for team data
Methods	`lm()`, `cor()`, `predict(..., interval = "prediction")`	Match formula to your response/predictor types
Results	Point prediction + **95% prediction interval** + CI for slope	These three numbers are the core deliverable
Conclusion	Interpret the PI with the sentence template from Read 2	*"I predict with 95% confidence that…"*

What you will do

1. Form a team (1–4 people); choose Option A (nesting dolls) or Option B (stone containers).

2. Agree on measurement rules; collect data in a shared Google Sheet or CSV.
3. Import data into R; make a scatterplot; compute `cor()`.
4. Fit `lm(response ~ predictor)`, interpret slope and intercept.
5. Use `predict(..., interval = "prediction")` for your target case.
6. Submit a **one-page PDF + runnable team notebook**.
7. Submit your **portfolio package** (both reference sheets + all five projects, polished).

Module 5 is four days

Day	Task
Mon Jun 29	Read 0 → Read 1
Tue Jun 30	Quiz A → Read 2 → Quiz B
Wed Jul 1	Demo lab → Exercise lab
Thu Jul 2	**Synthesis quiz · Milestone · Project 5 · Portfolio package — all due**

⚠ Everything due Thursday July 2

Module 5 is compressed. There is **no Sunday extension** for Project 5. Plan to finish the exercise lab Wednesday so Thursday is for synthesis and polishing your portfolio.

What to do next

1. Open Project 5 and choose Option A or B with your team (5 minutes).
2. Confirm your team in the Canvas Module 5 milestone.
3. Start **Read 1** — correlation through residuals and model performance.

When you finish Read 1, take Quiz A (Def & Notation), then continue with Read 2.